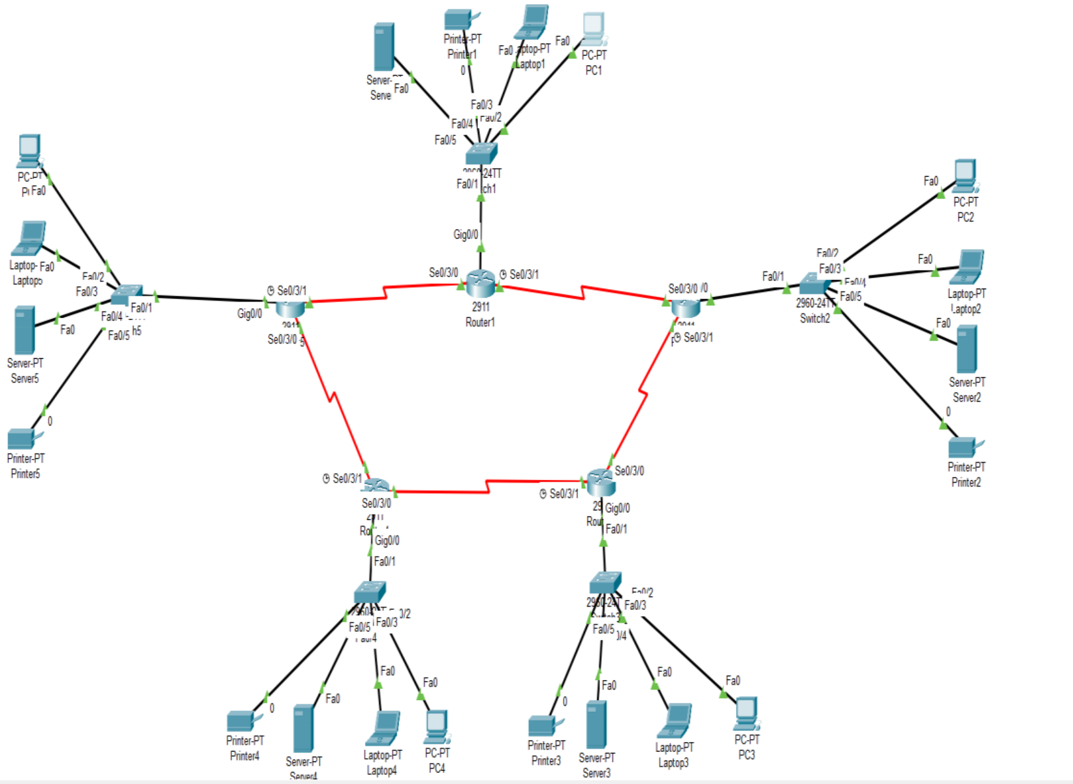
CCNA Capstone Project: Global Logistics Network

1-Network Topology

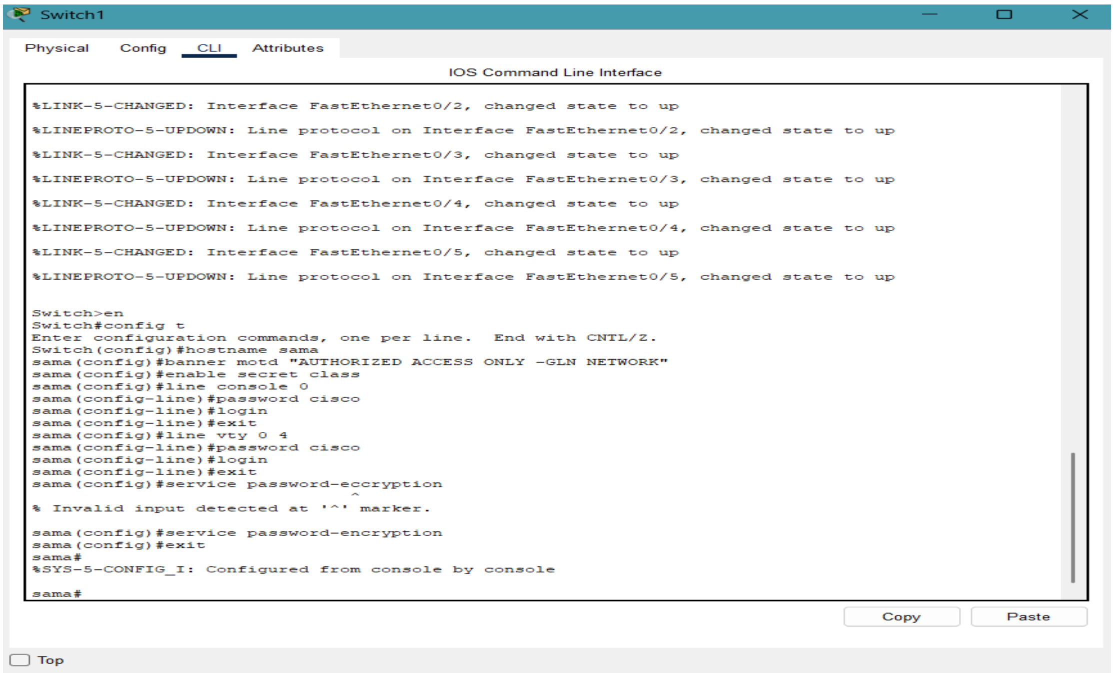
This topology represents a redundant ring network connecting 5 Branches using EIGRP as the dynamic routing protocol.



2-Switch configurations:

In this section, I performed the initial device hardening. This includes setting a hostname, securing the console and remote access with passwords, and enabling global password encryption to protect sensitive data.

I implemented VLAN (Virtual Local Area Network) technology to logically segment the network into different departments. This improves security and reduces broadcast traffic, ensuring that each department stays within its own broadcast domain.



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-eccryption
                                 ^
% Invalid input detected at '^' marker.

sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#
```

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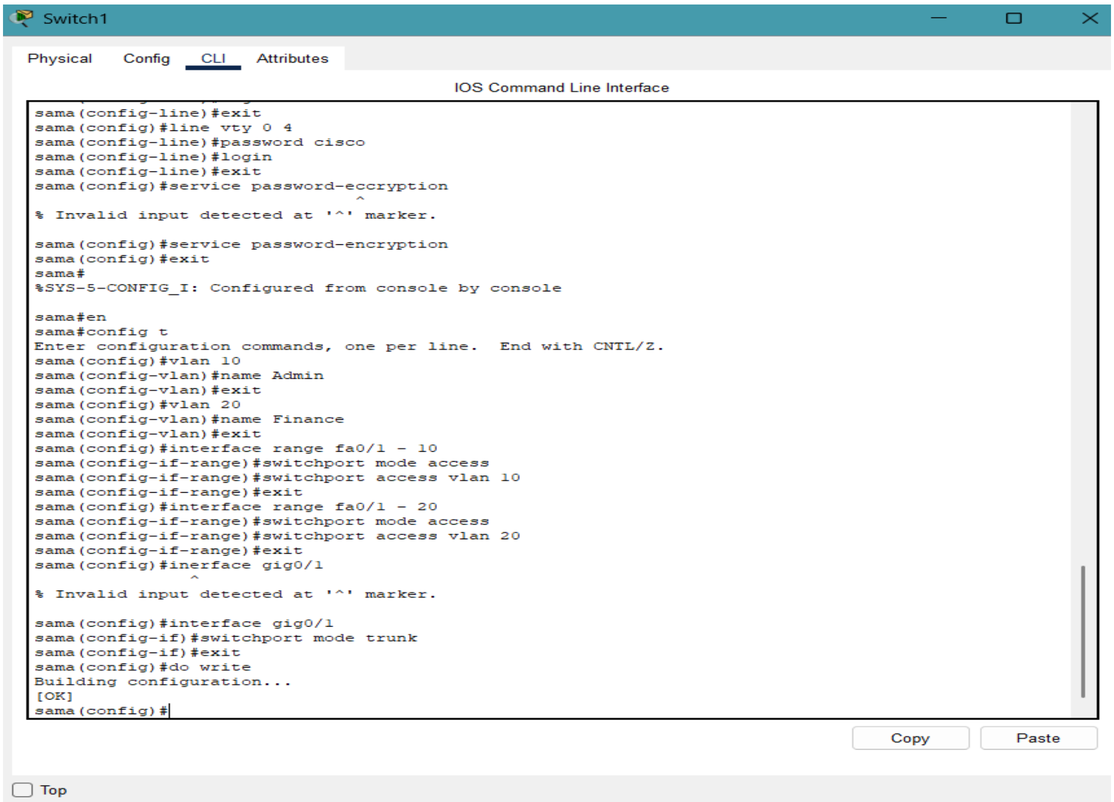
Top



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-eccryption
                                 ^
% Invalid input detected at '^' marker.

sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#en
sama#config t
Enter configuration commands, one per line.  End with CNTL/Z.
sama(config)#vlan 10
sama(config-vlan)#name Admin
sama(config-vlan)#exit
sama(config)#vlan 20
sama(config-vlan)#name Finance
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 10
sama(config-if-range)#exit
sama(config)#interface range fa0/1 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 20
sama(config-if-range)#exit
sama(config)#inerface gig0/1
                ^
% Invalid input detected at '^' marker.

sama(config)#interface gig0/1
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```

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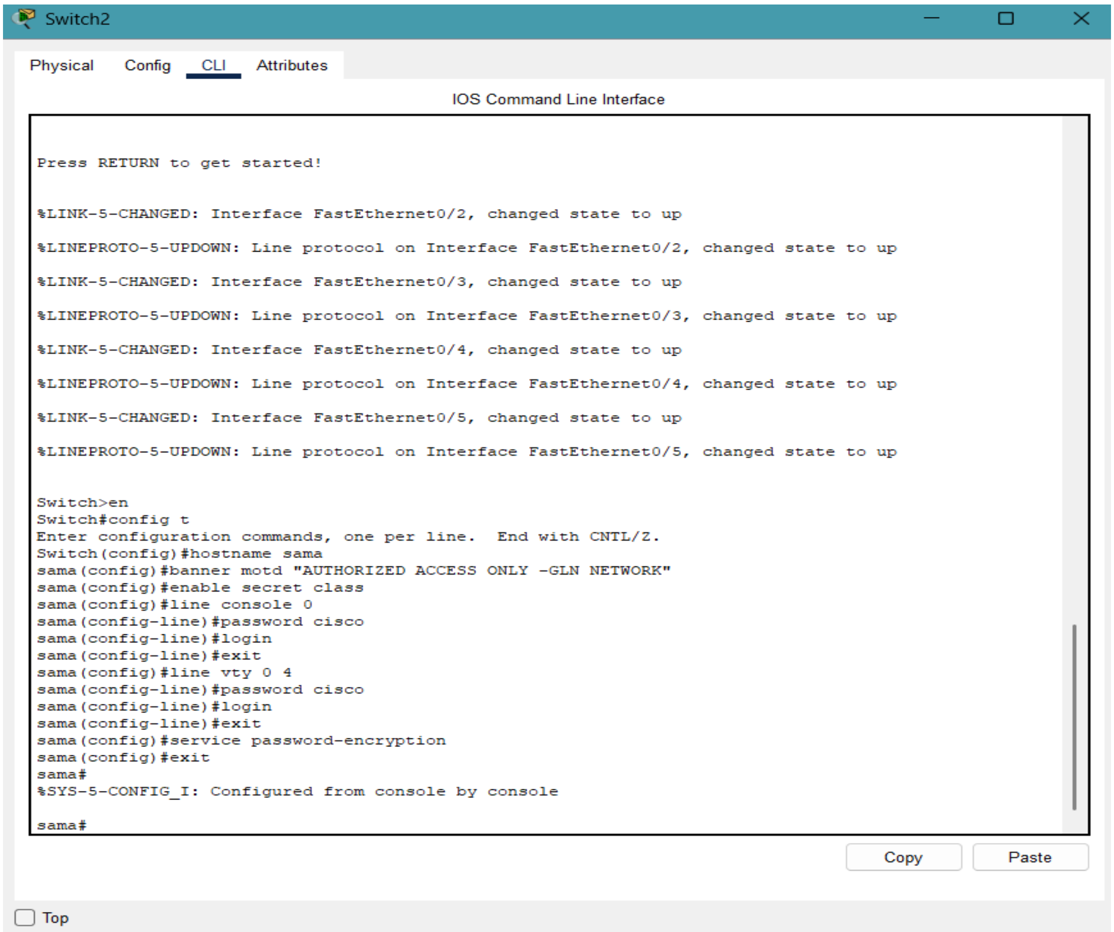
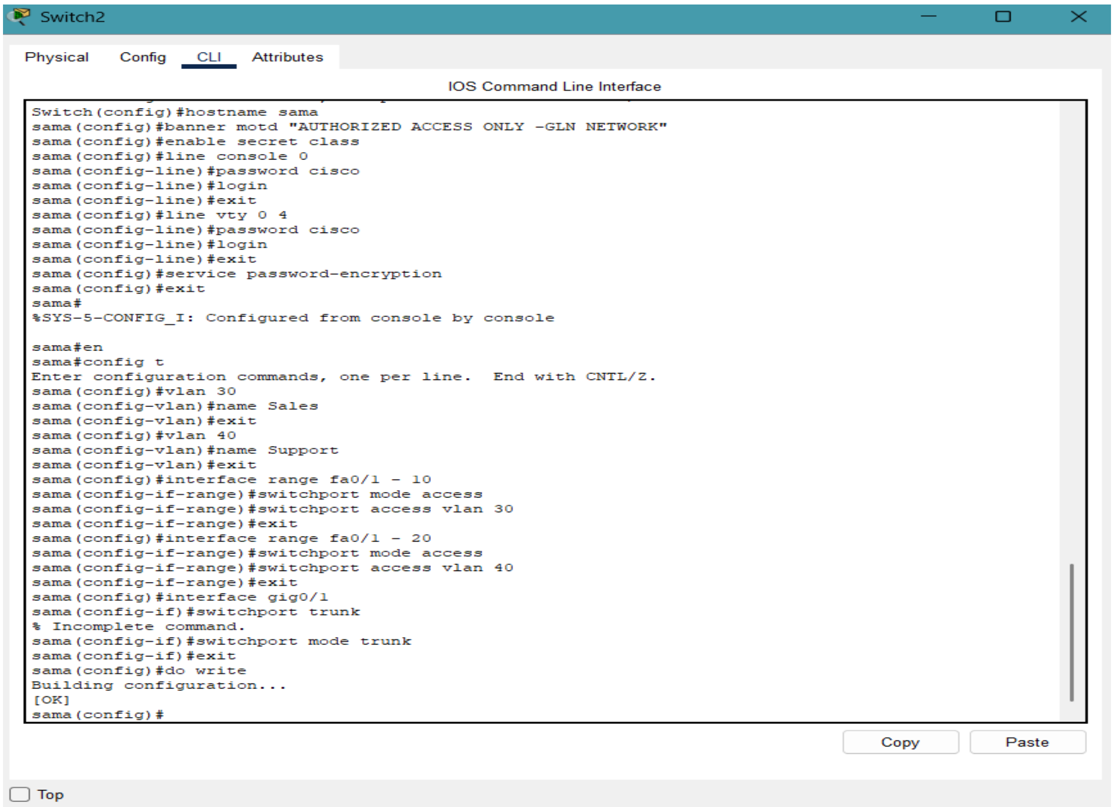
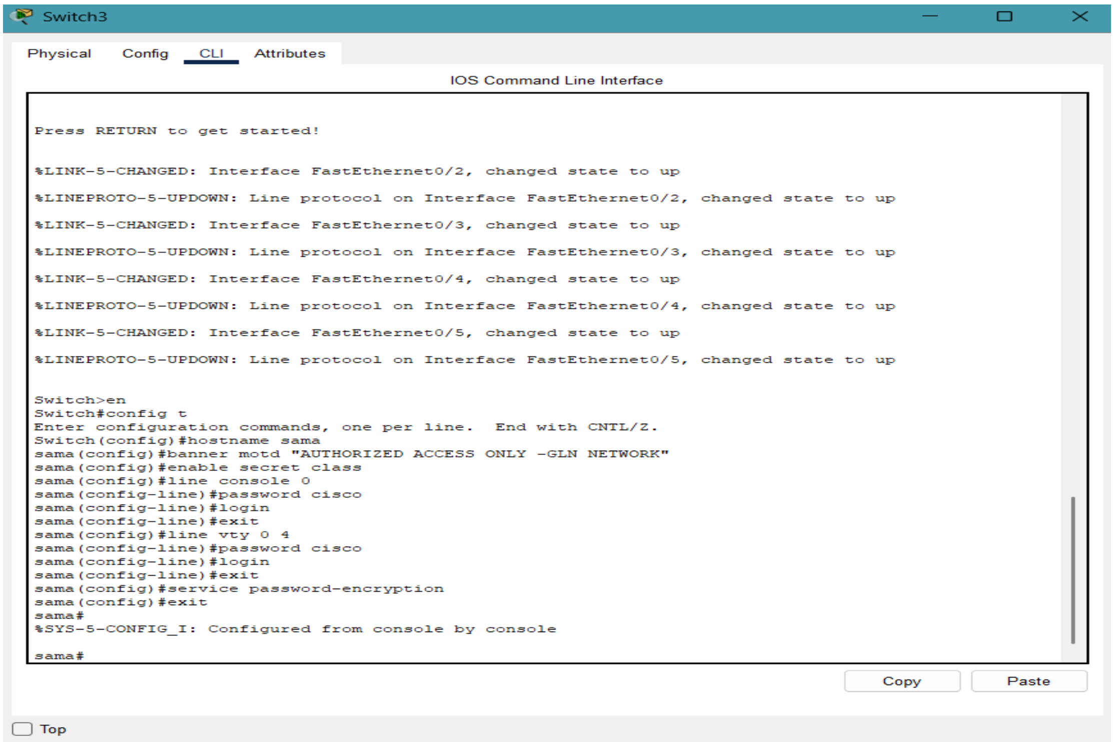
Top


Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#
Copy
Paste
Top


Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#en
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#vlan 30
sama(config-vlan)#name Sales
sama(config-vlan)#exit
sama(config)#vlan 40
sama(config-vlan)#name Support
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 30
sama(config-if-range)#exit
sama(config)#interface range fa0/1 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 40
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport trunk
% Incomplete command.
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top


Switch3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#
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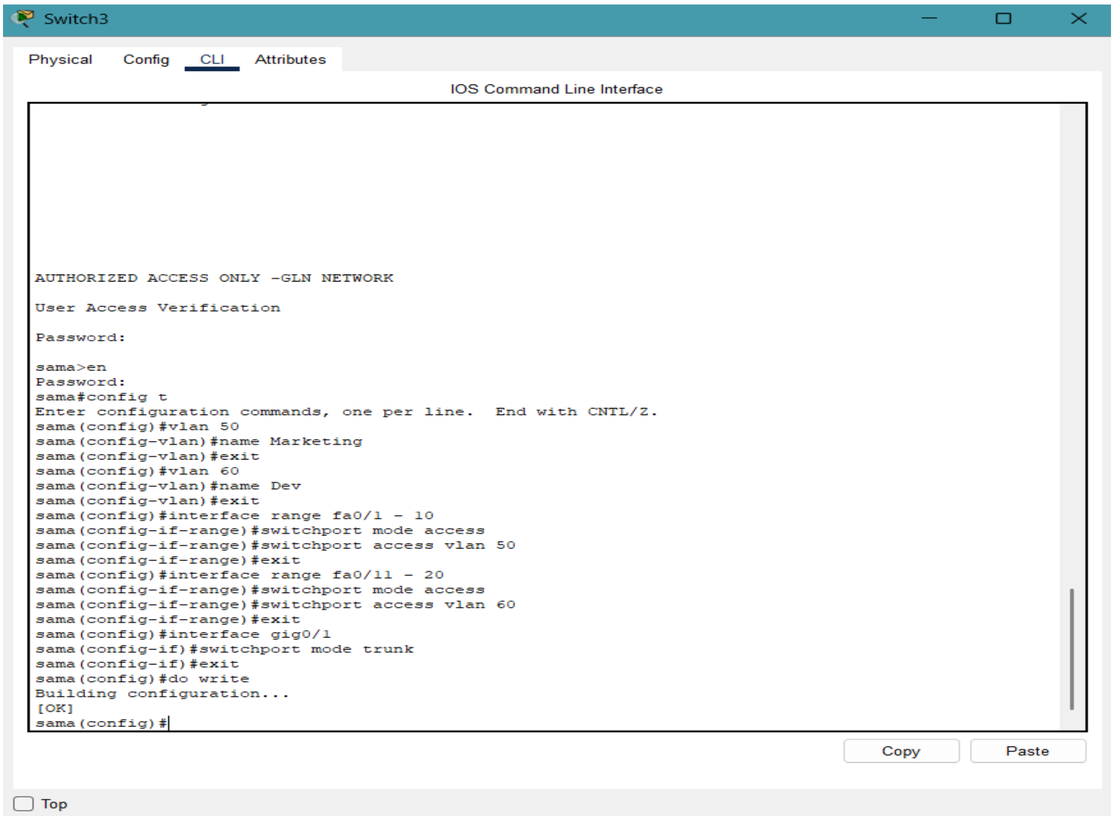
Top


Switch3
Physical
Config
CLI
Attributes
IOS Command Line Interface
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#vlan 50
sama(config-vlan)#name Marketing
sama(config-vlan)#exit
sama(config)#vlan 60
sama(config-vlan)#name Dev
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 50
sama(config-if-range)#exit
sama(config)#interface range fa0/11 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 60
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top

Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#
```

Copy Paste

Top

Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.

AUTHORIZED ACCESS ONLY -GLN NETWORK

User Access Verification

Password:

sama>en
sama#config t
Enter configuration commands, one per line.  End with CNTL/Z.
sama(config)#vlan 70
sama(config-vlan)#name Inventory
sama(config-vlan)#exit
sama(config)#vlan 80
sama(config-vlan)#name Shipping
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 70
sama(config-if-range)#exit
sama(config)#interface range fa0/11 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 80
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```

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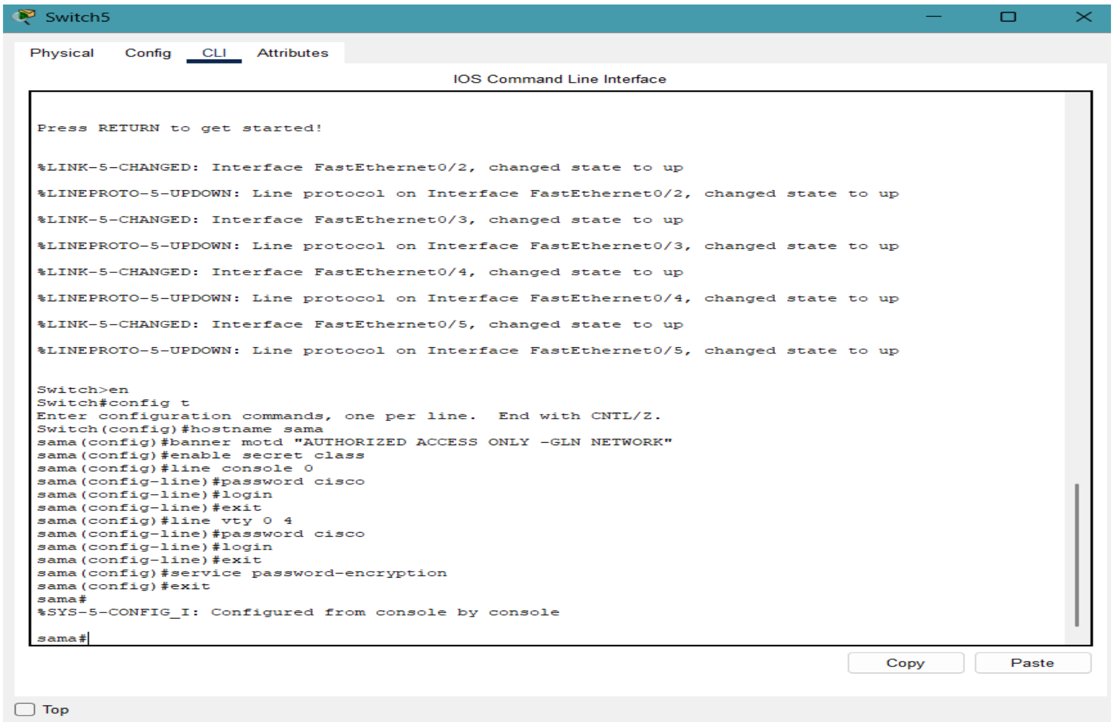
Switch5
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#
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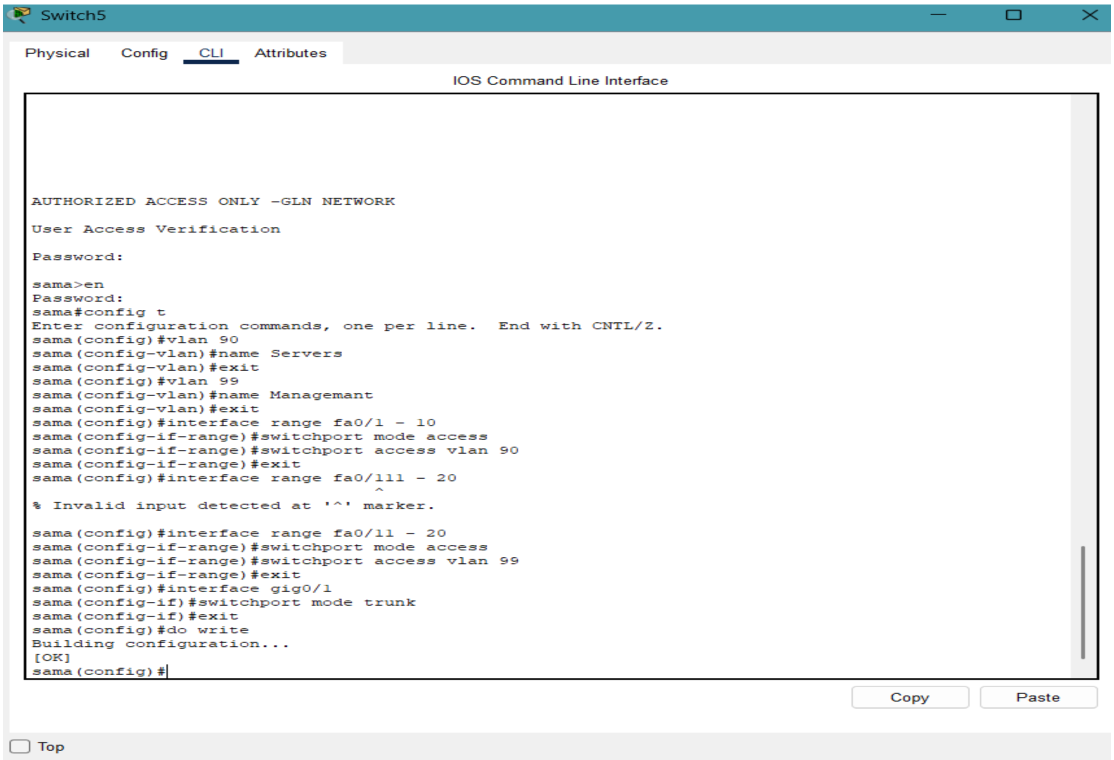
Top


Switch5
Physical
Config
CLI
Attributes
IOS Command Line Interface
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#vlan 90
sama(config-vlan)#name Servers
sama(config-vlan)#exit
sama(config)#vlan 99
sama(config-vlan)#name Managemant
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 90
sama(config-if-range)#exit
sama(config)#interface range fa0/111 - 20
^
% Invalid input detected at '^' marker.
sama(config)#interface range fa0/11 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 99
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
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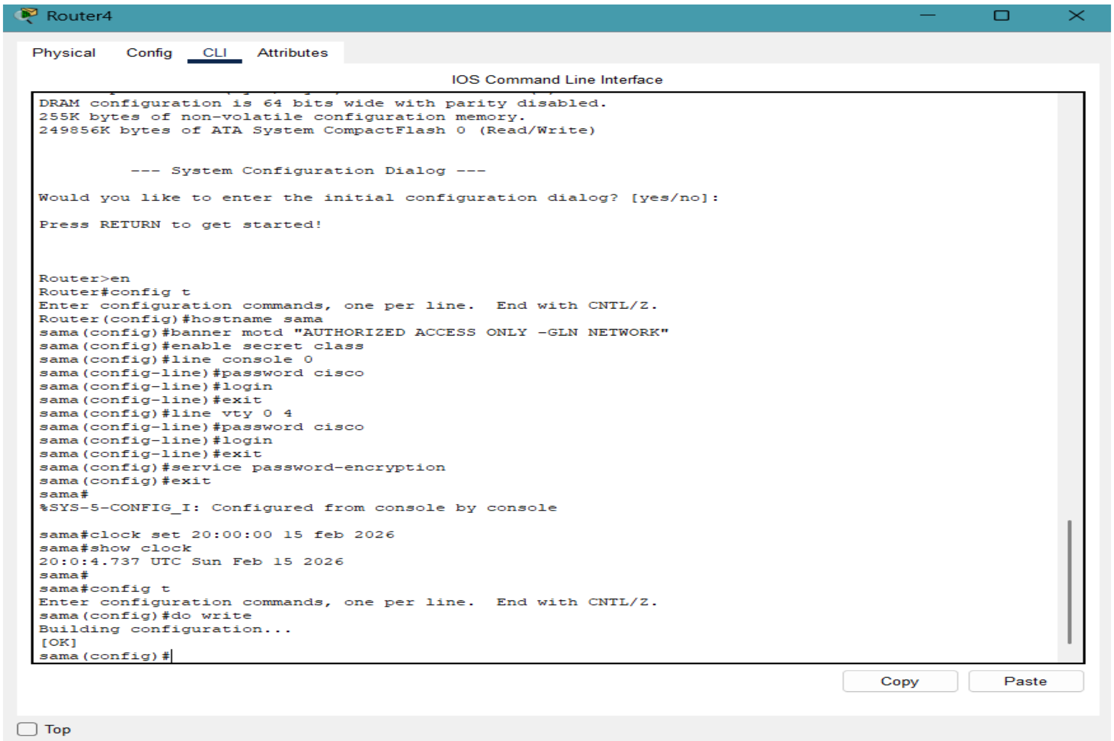
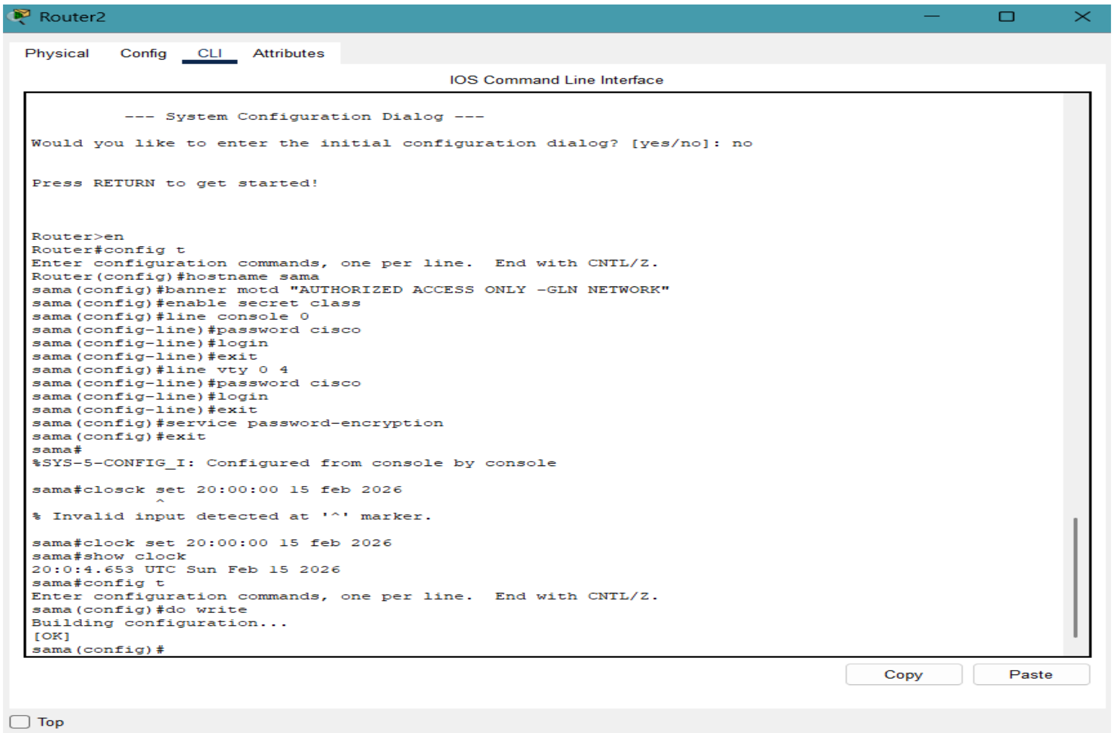
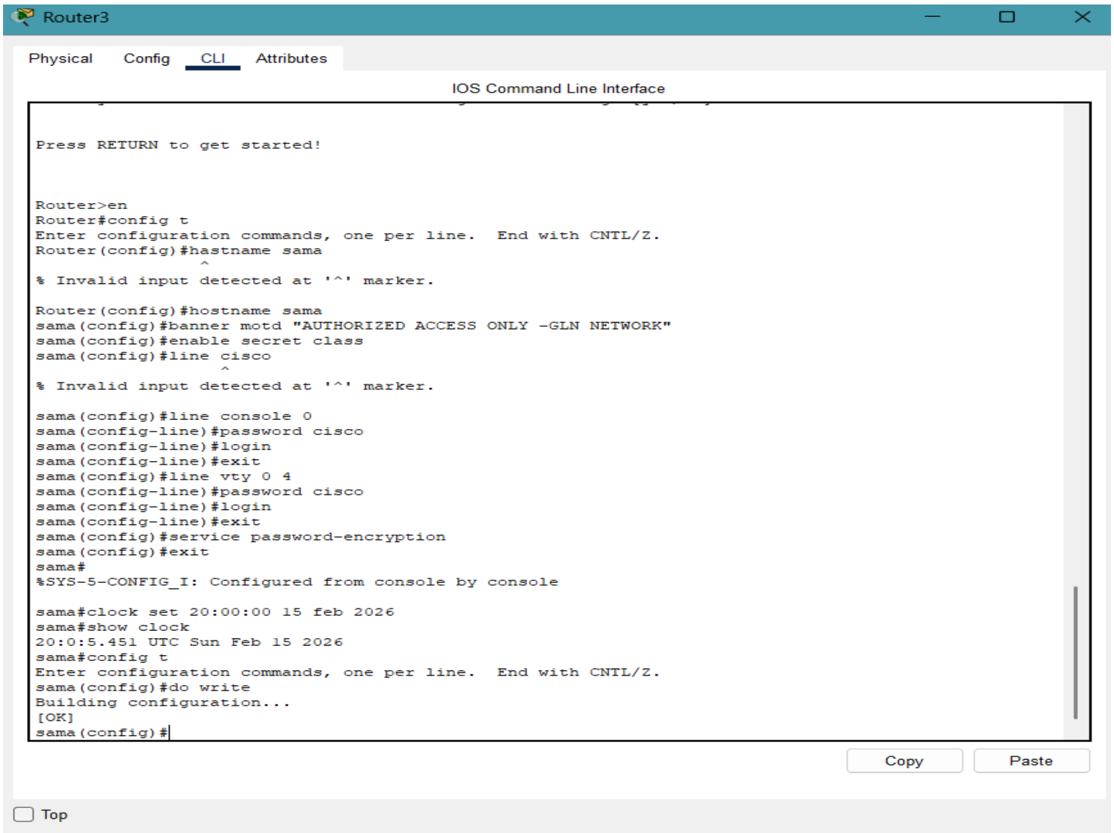
Top

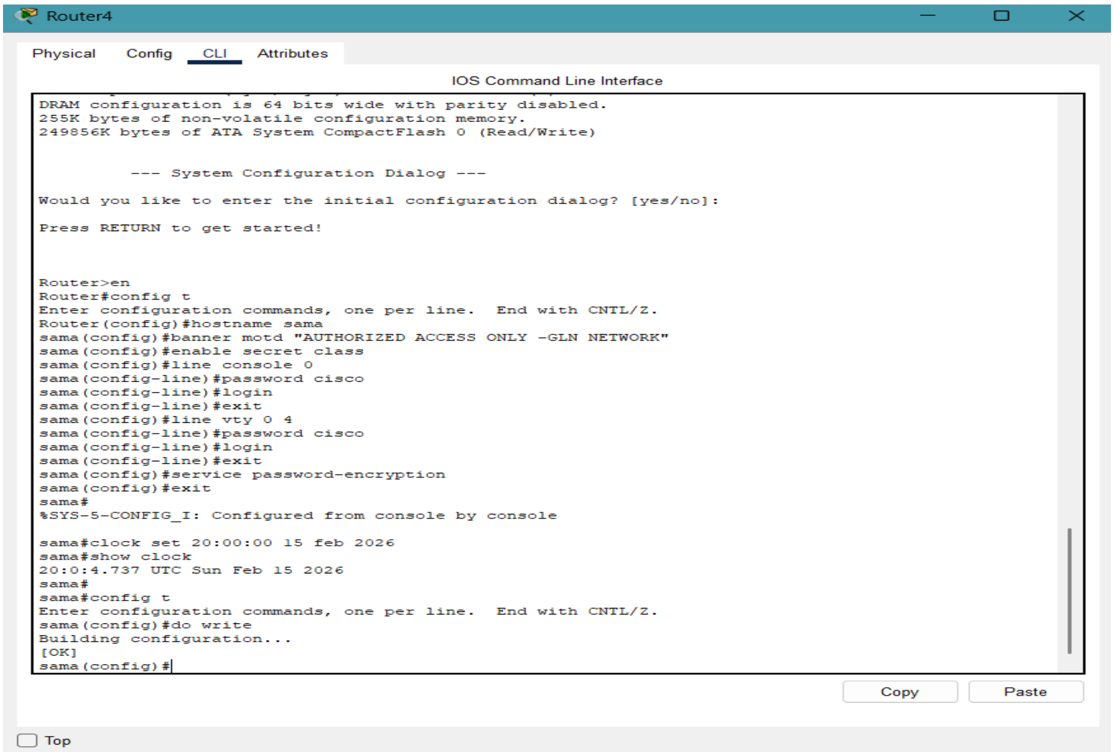
3-Router Configurations:

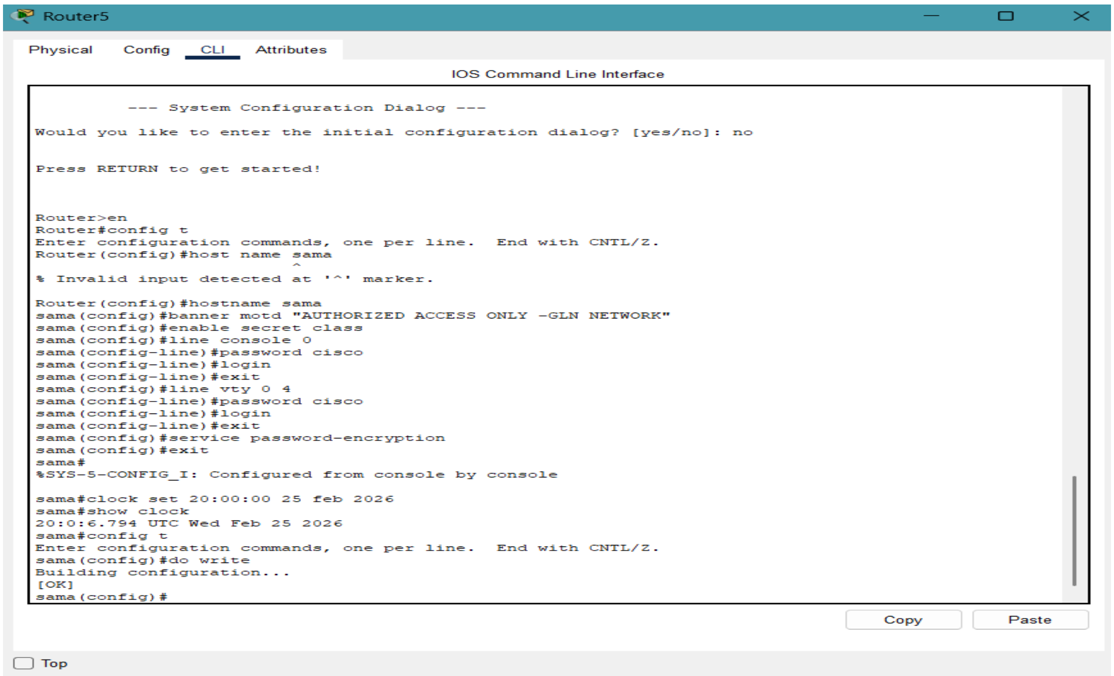
In this section, I performed the initial device hardening. This includes setting a hostname, securing the console and remote access with passwords, and enabling global password encryption to protect sensitive data.





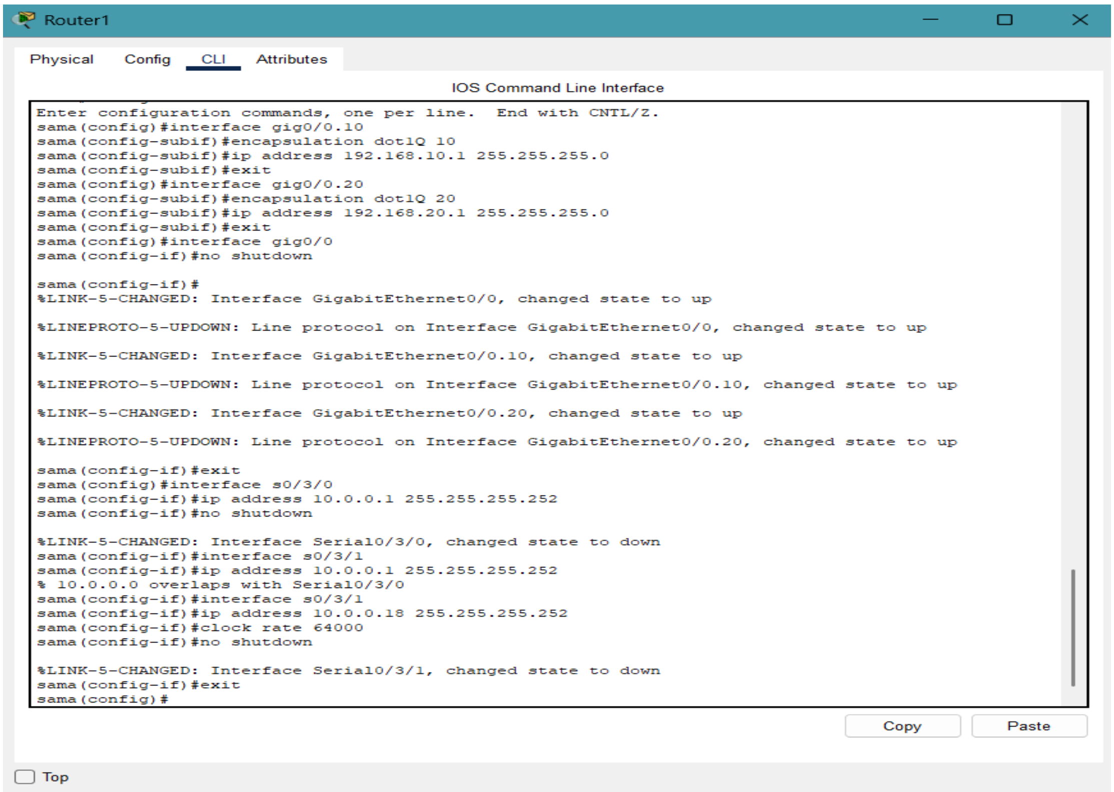

Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hastname sama
% Invalid input detected at '^' marker.
Router(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line cisco
% Invalid input detected at '^' marker.
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#clock set 20:00:00 15 feb 2026
sama#show clock
20:0:5.451 UTC Sun Feb 15 2026
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top

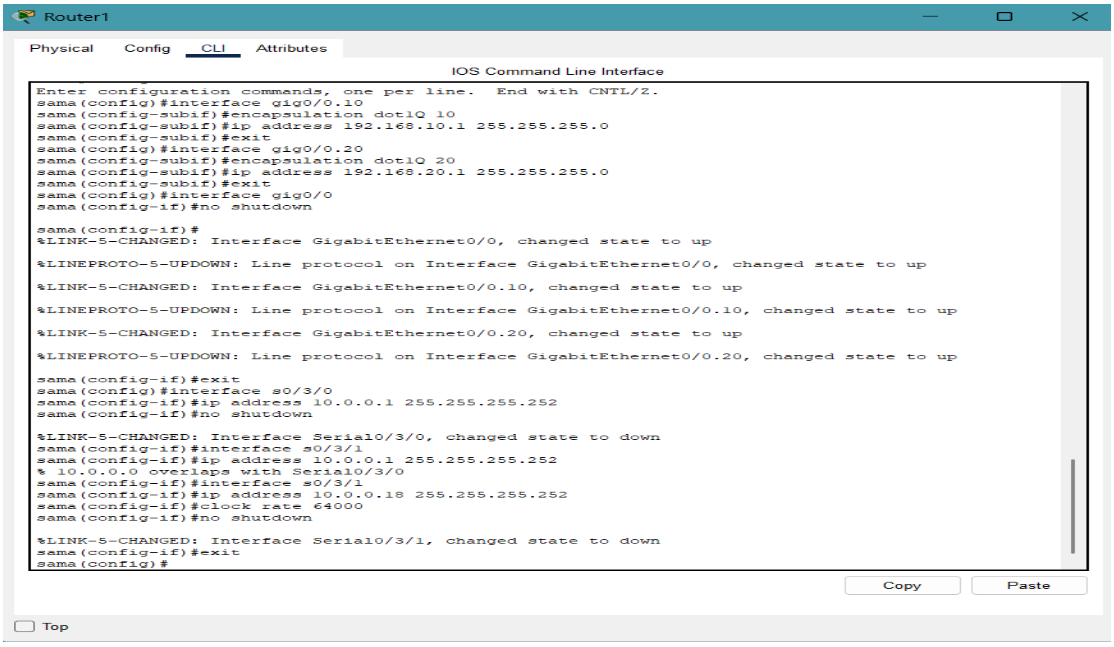

Router4
Physical
Config
CLI
Attributes
IOS Command Line Interface
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#clock set 20:00:00 15 feb 2026
sama#show clock
20:0:4.737 UTC Sun Feb 15 2026
sama#
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top



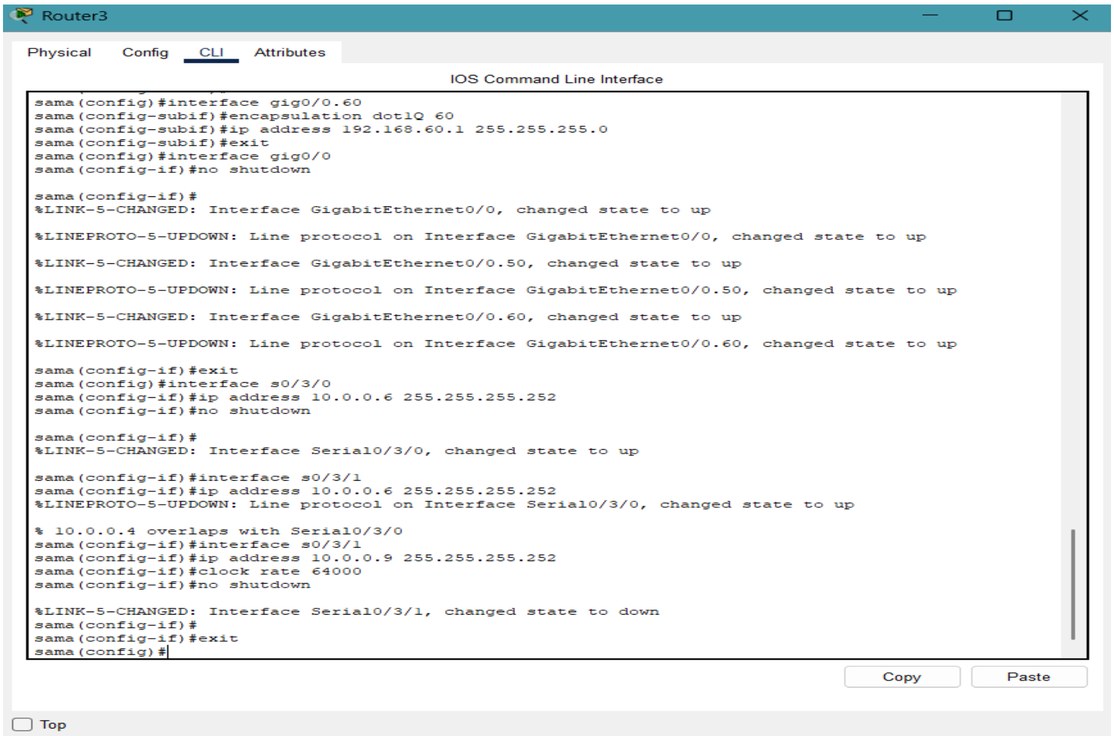
After

creating the VLANs, I assigned specific ports to their respective VLANs in Access mode. Additionally, I configured the Gigabit ports as Trunk links to allow multiple VLANs to communicate across the entire network infrastructure.

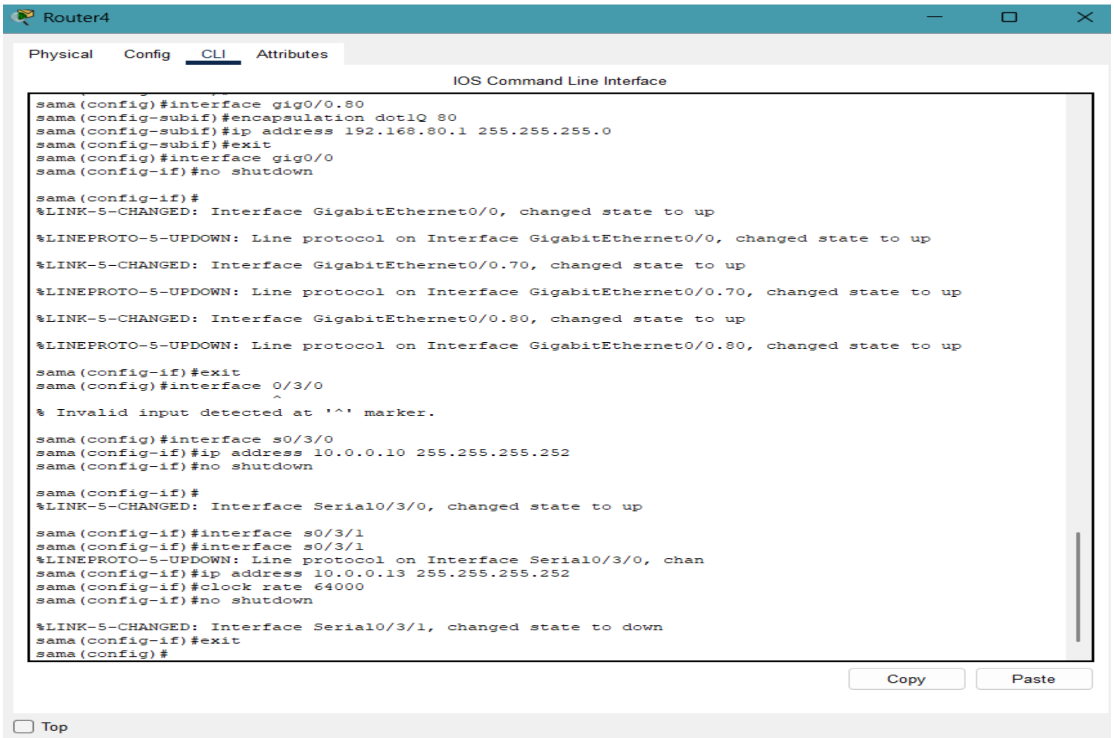





```
Enter configuration commands, one per line.  End with CNTL/Z.
sama(config)#interface gig0/0.10
sama(config-subif)#encapsulation dot1Q 10
sama(config-subif)#ip address 192.168.10.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0.20
sama(config-subif)#encapsulation dot1Q 20
sama(config-subif)#ip address 192.168.20.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

sama(config-if)#exit
sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.1 255.255.255.252
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
sama(config-if)#interface s0/3/1
sama(config-if)#ip address 10.0.0.1 255.255.255.252
% 10.0.0.0 overlaps with Serial0/3/0
sama(config-if)#interface s0/3/1
sama(config-if)#ip address 10.0.0.18 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#exit
sama(config)#
```




```
sama(config)#interface gig0/0.60
sama(config-subif)#encapsulation dot1Q 60
sama(config-subif)#ip address 192.168.60.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.60, changed state to up

sama(config-if)#exit
sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.6 255.255.255.252
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

sama(config-if)#interface s0/3/1
sama(config-if)#ip address 10.0.0.6 255.255.255.252
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

% 10.0.0.4 overlaps with Serial0/3/0
sama(config-if)#interface s0/3/1
sama(config-if)#ip address 10.0.0.9 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#
sama(config-if)#exit
sama(config)#
```


Router4
Physical
Config
CLI
Attributes
IOS Command Line Interface
sama(config)#interface gig0/0.80
sama(config-subif)#encapsulation dot1Q 80
sama(config-subif)#ip address 192.168.80.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0
sama(config-if)#no shutdown
sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.70, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.70, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.80, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.80, changed state to up
sama(config-if)#exit
sama(config)#interface 0/3/0
^
% Invalid input detected at '^' marker.
sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.10 255.255.255.252
sama(config-if)#no shutdown
sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
sama(config-if)#interface s0/3/1
sama(config-if)#interface s0/3/1
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, chan
sama(config-if)#ip address 10.0.0.13 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#exit
sama(config)#
Copy
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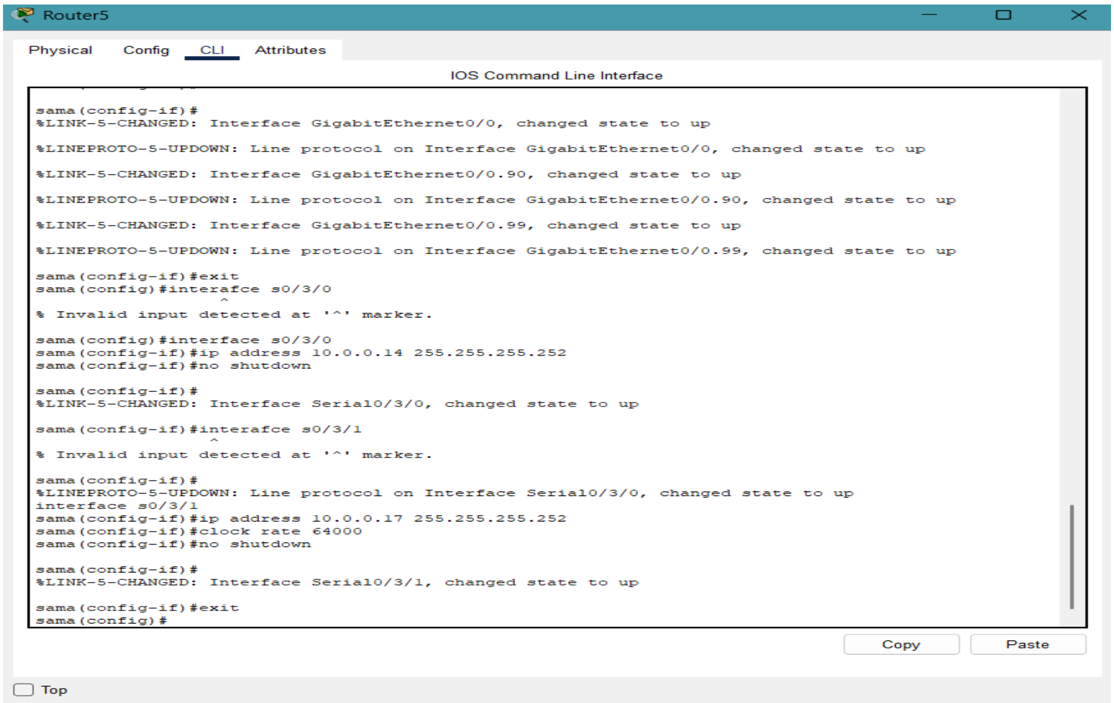
Top


Router5
Physical
Config
CLI
Attributes
IOS Command Line Interface
sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.90, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up
sama(config-if)#exit
sama(config)#interafce s0/3/0
^
% Invalid input detected at '^' marker.
sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.14 255.255.255.252
sama(config-if)#no shutdown
sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
sama(config-if)#interafce s0/3/1
^
% Invalid input detected at '^' marker.
sama(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
interface s0/3/1
sama(config-if)#ip address 10.0.0.17 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown
sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
sama(config-if)#exit
sama(config)#
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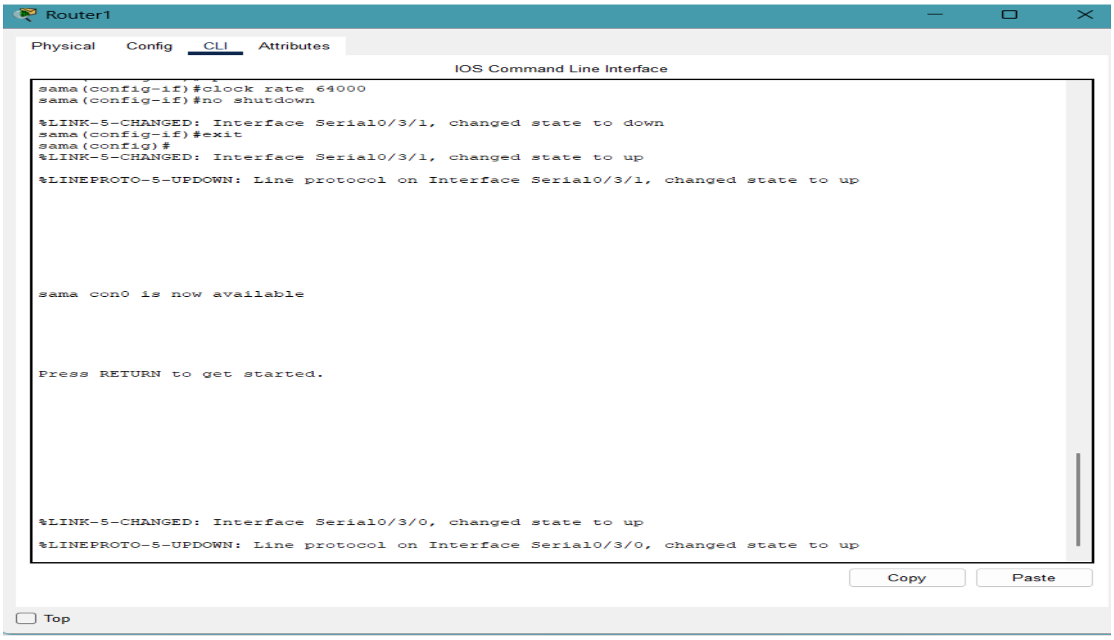
Top


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#exit
sama(config)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
sama con0 is now available
Press RETURN to get started.
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
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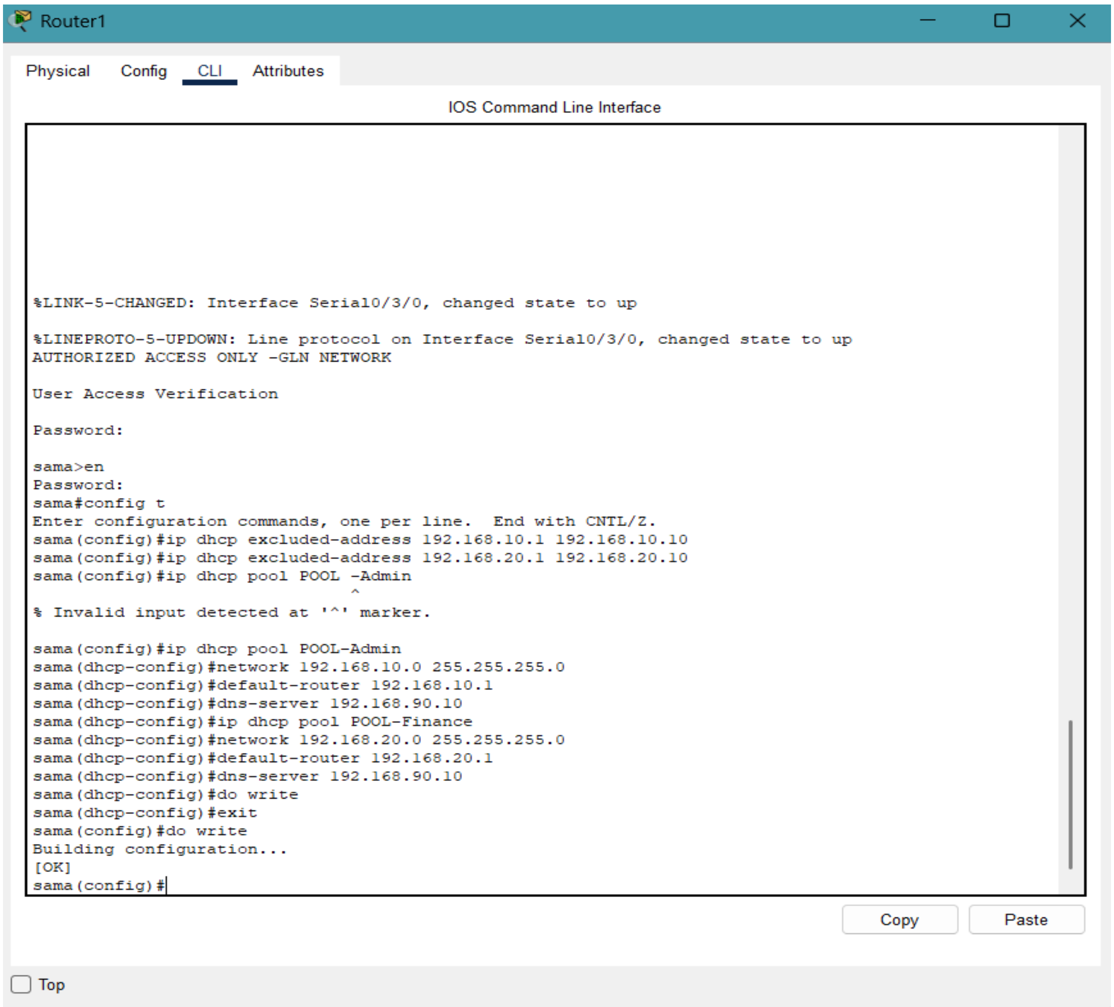
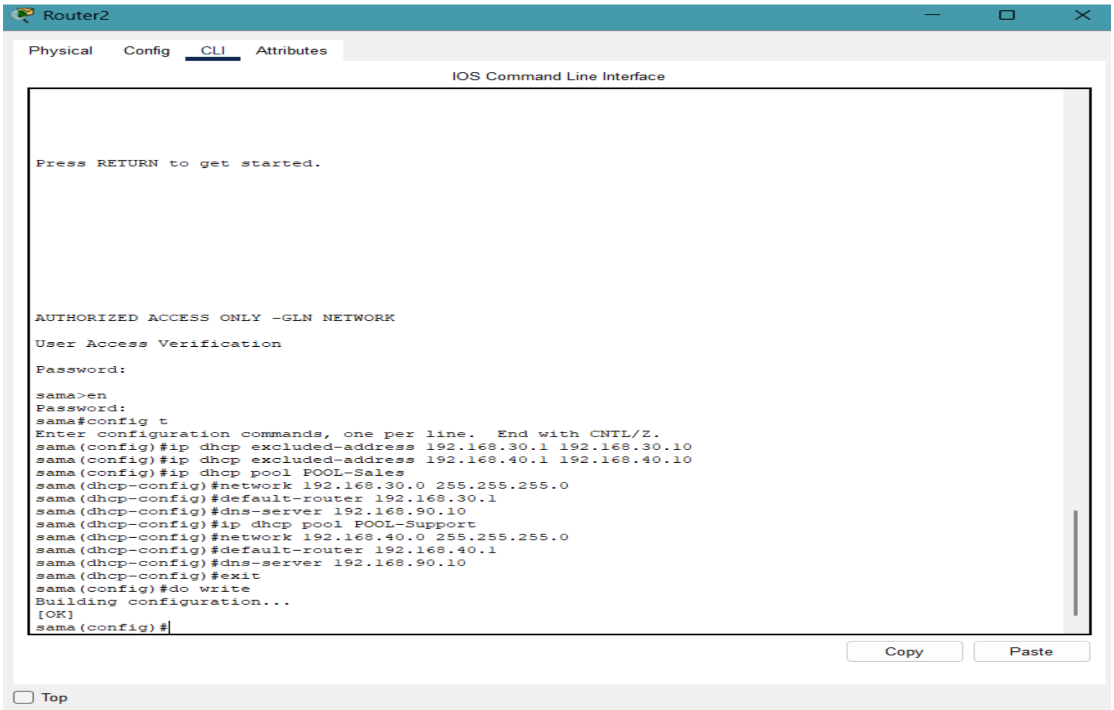
Top


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
sama(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
sama(config)#ip dhcp pool POOL -Admin
^
% Invalid input detected at '^' marker.
sama(config)#ip dhcp pool POOL-Admin
sama(dhcp-config)#network 192.168.10.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.10.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#ip dhcp pool POOL-Finance
sama(dhcp-config)#network 192.168.20.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.20.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#do write
sama(dhcp-config)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
sama(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.10
sama(config)#ip dhcp pool POOL-Sales
sama(dhcp-config)#network 192.168.30.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.30.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#ip dhcp pool POOL-Support
sama(dhcp-config)#network 192.168.40.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.40.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
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Paste
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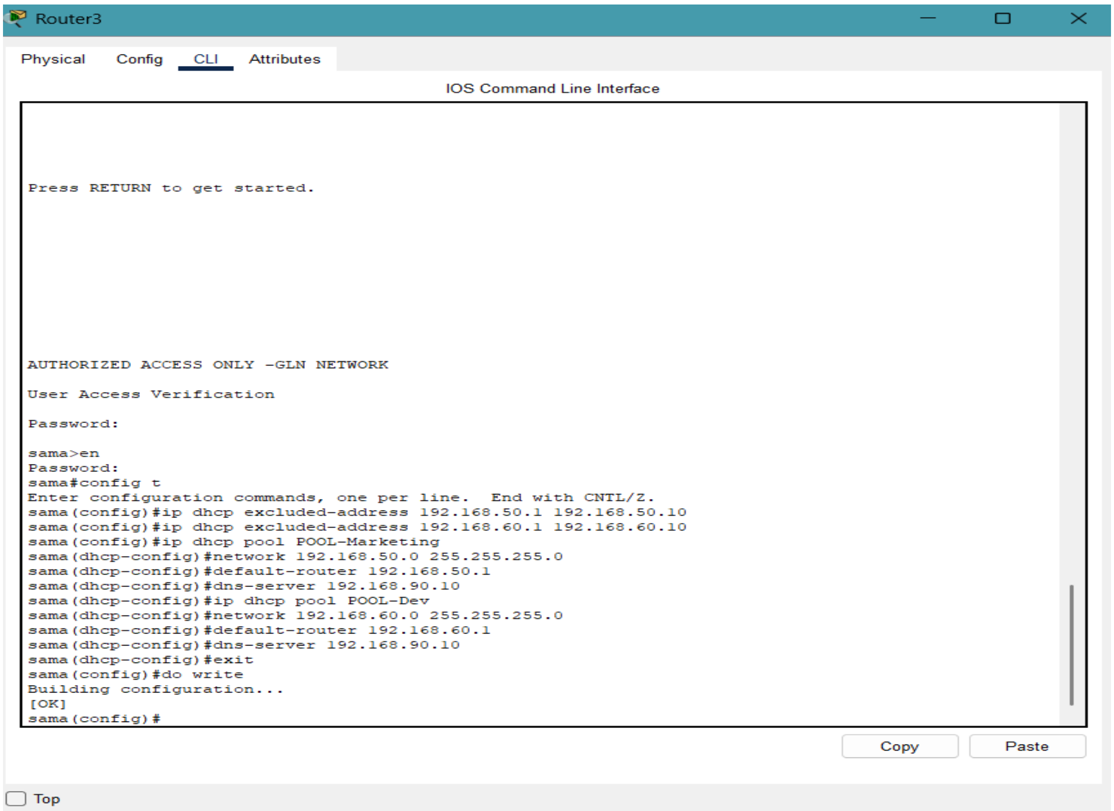
Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#ip dhcp excluded-address 192.168.50.1 192.168.50.10
sama(config)#ip dhcp excluded-address 192.168.60.1 192.168.60.10
sama(config)#ip dhcp pool POOL-Marketing
sama(dhcp-config)#network 192.168.50.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.50.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#ip dhcp pool POOL-Dev
sama(dhcp-config)#network 192.168.60.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.60.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
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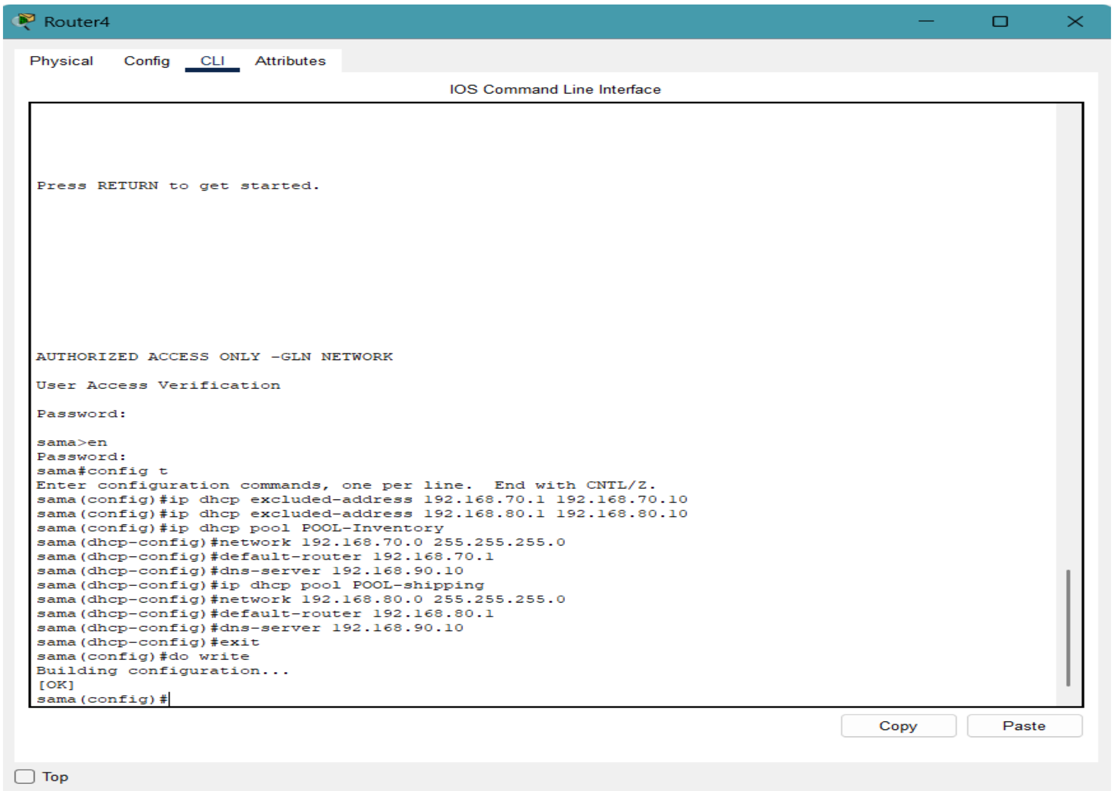
Router4
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#ip dhcp excluded-address 192.168.70.1 192.168.70.10
sama(config)#ip dhcp excluded-address 192.168.80.1 192.168.80.10
sama(config)#ip dhcp pool POOL-Inventory
sama(dhcp-config)#network 192.168.70.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.70.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#ip dhcp pool POOL-shipping
sama(dhcp-config)#network 192.168.80.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.80.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
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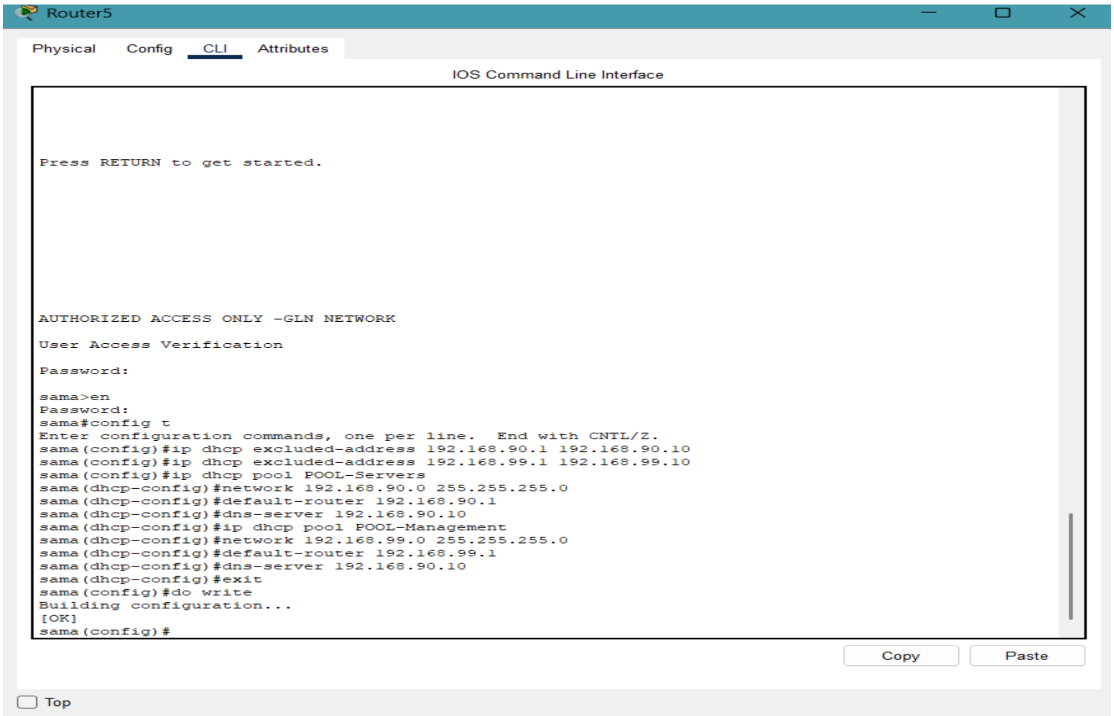
Router5
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#ip dhcp excluded-address 192.168.90.1 192.168.90.10
sama(config)#ip dhcp excluded-address 192.168.99.1 192.168.99.10
sama(config)#ip dhcp pool POOL-Servers
sama(dhcp-config)#network 192.168.90.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.90.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#ip dhcp pool POOL-Management
sama(dhcp-config)#network 192.168.99.0 255.255.255.0
sama(dhcp-config)#default-router 192.168.99.1
sama(dhcp-config)#dns-server 192.168.90.10
sama(dhcp-config)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
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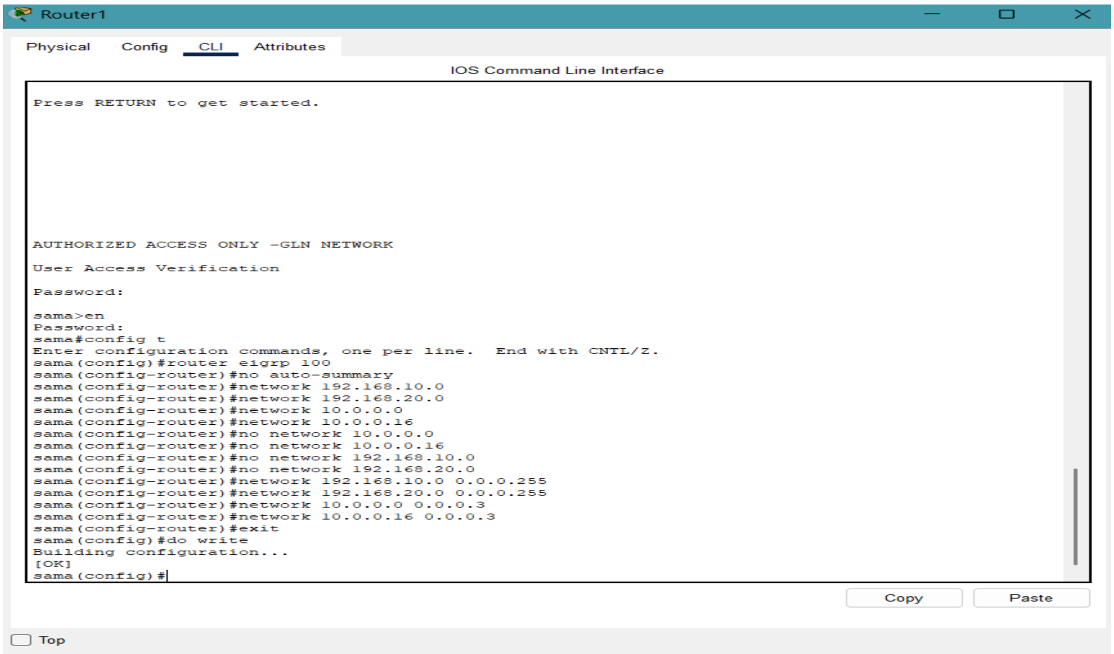
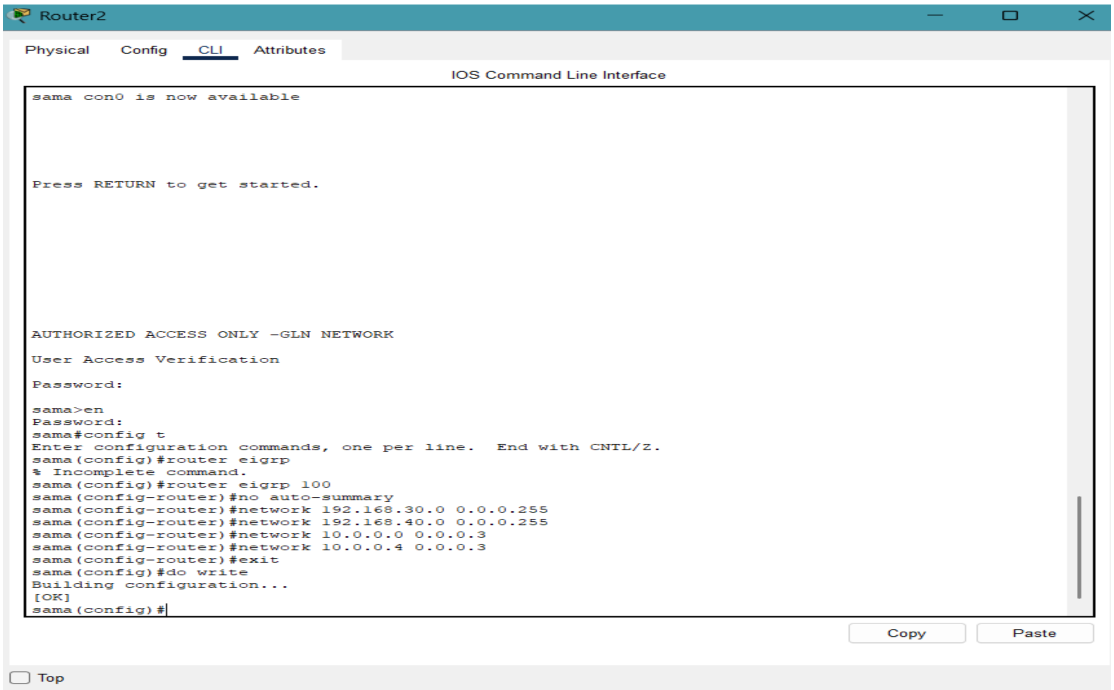
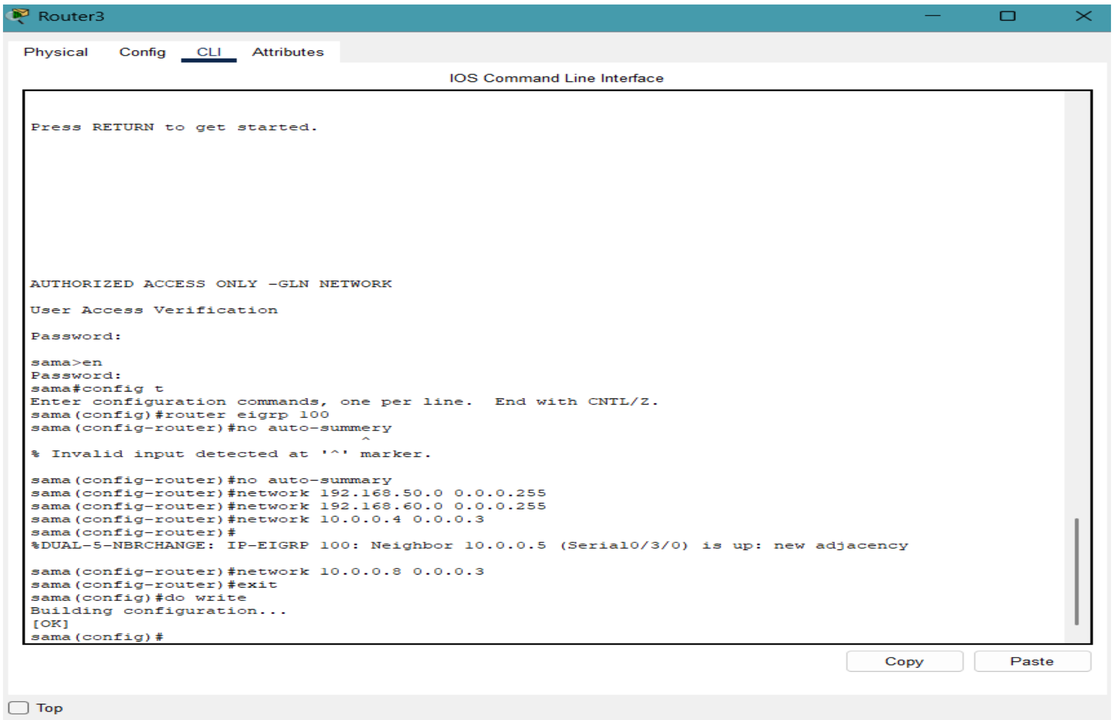

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.10.0
sama(config-router)#network 192.168.20.0
sama(config-router)#network 10.0.0.0
sama(config-router)#network 10.0.0.16
sama(config-router)#no network 10.0.0.0
sama(config-router)#no network 10.0.0.16
sama(config-router)#no network 192.168.10.0
sama(config-router)#no network 192.168.20.0
sama(config-router)#network 192.168.10.0 0.0.0.255
sama(config-router)#network 192.168.20.0 0.0.0.255
sama(config-router)#network 10.0.0.0 0.0.0.3
sama(config-router)#network 10.0.0.16 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
sama con0 is now available
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp
% Incomplete command.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.30.0 0.0.0.255
sama(config-router)#network 192.168.40.0 0.0.0.255
sama(config-router)#network 10.0.0.0 0.0.0.3
sama(config-router)#network 10.0.0.4 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
Paste
Top


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summery
^
% Invalid input detected at '^' marker.
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.50.0 0.0.0.255
sama(config-router)#network 192.168.60.0 0.0.0.255
sama(config-router)#network 10.0.0.4 0.0.0.3
sama(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.5 (Serial0/3/0) is up: new adjacency
sama(config-router)#network 10.0.0.8 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
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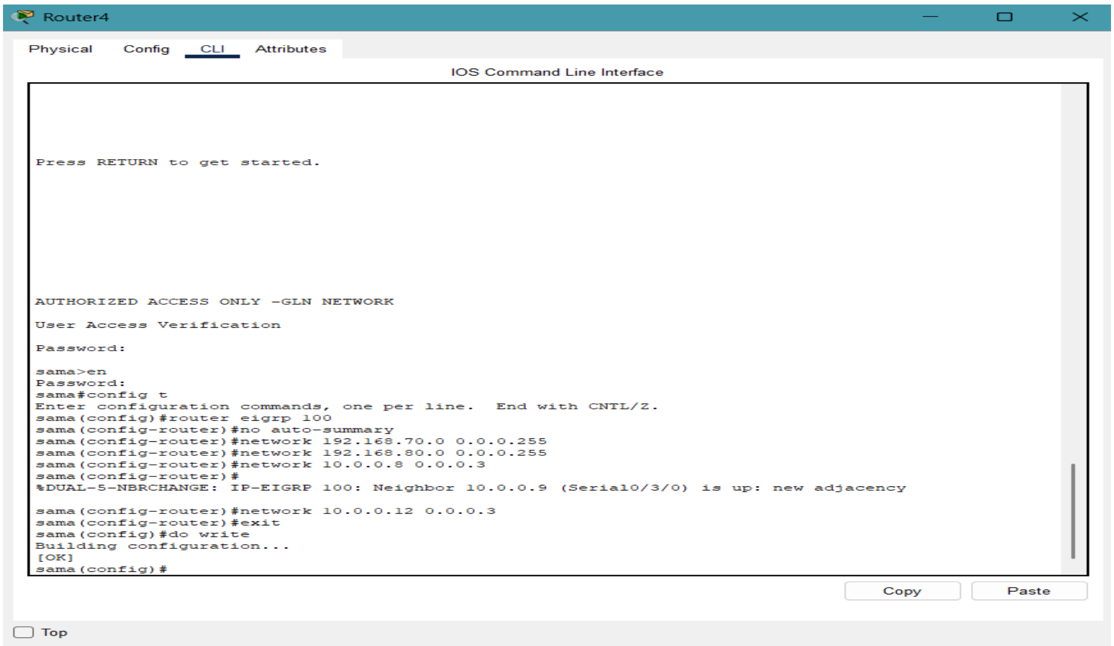
Top


Router4
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
AUTHORIZED ACCESS ONLY -GLN NETWORK
User Access Verification
Password:
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.70.0 0.0.0.255
sama(config-router)#network 192.168.80.0 0.0.0.255
sama(config-router)#network 10.0.0.8 0.0.0.3
sama(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.9 (Serial0/3/0) is up: new adjacency
sama(config-router)#network 10.0.0.12 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#
Copy
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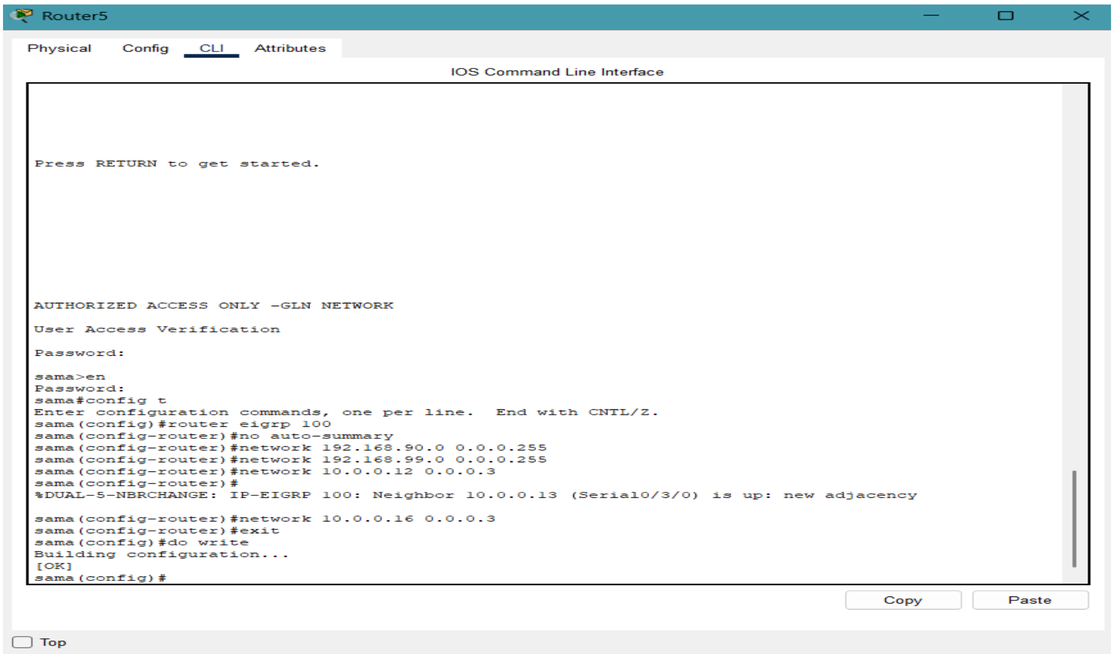
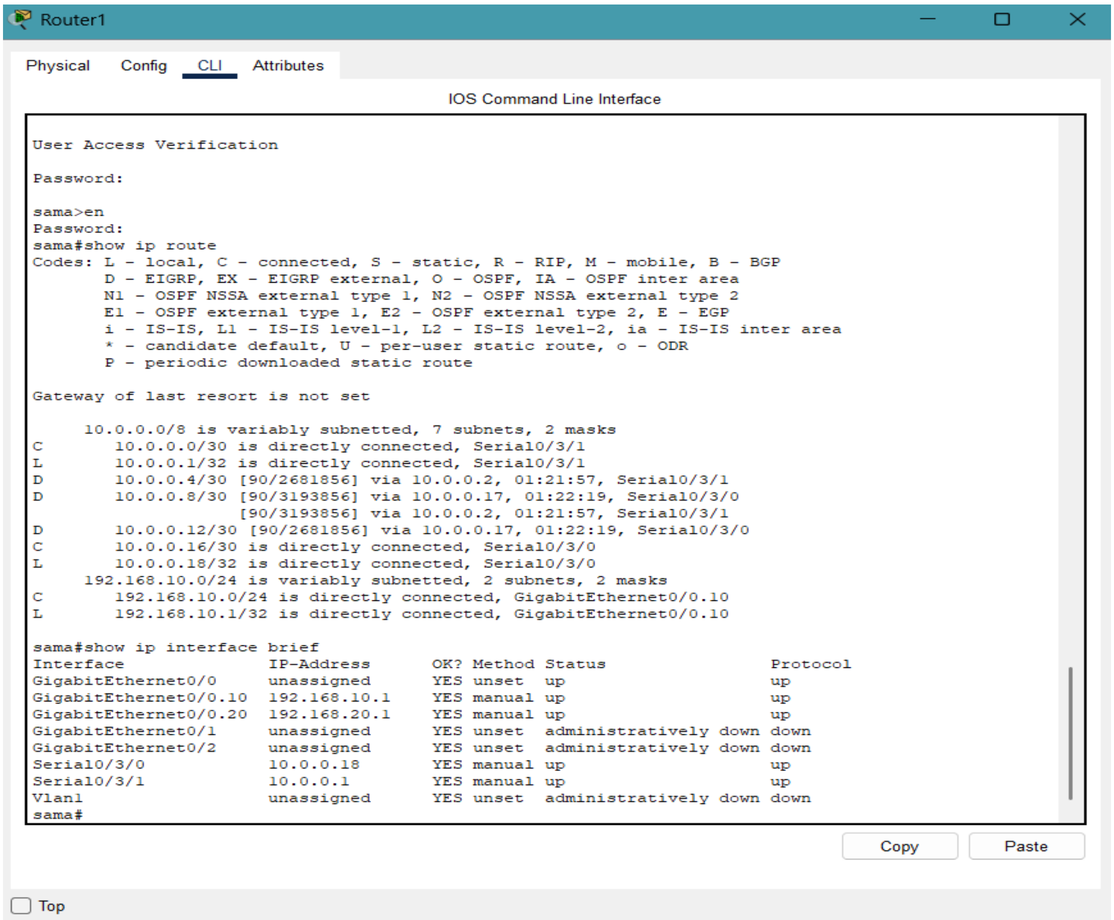
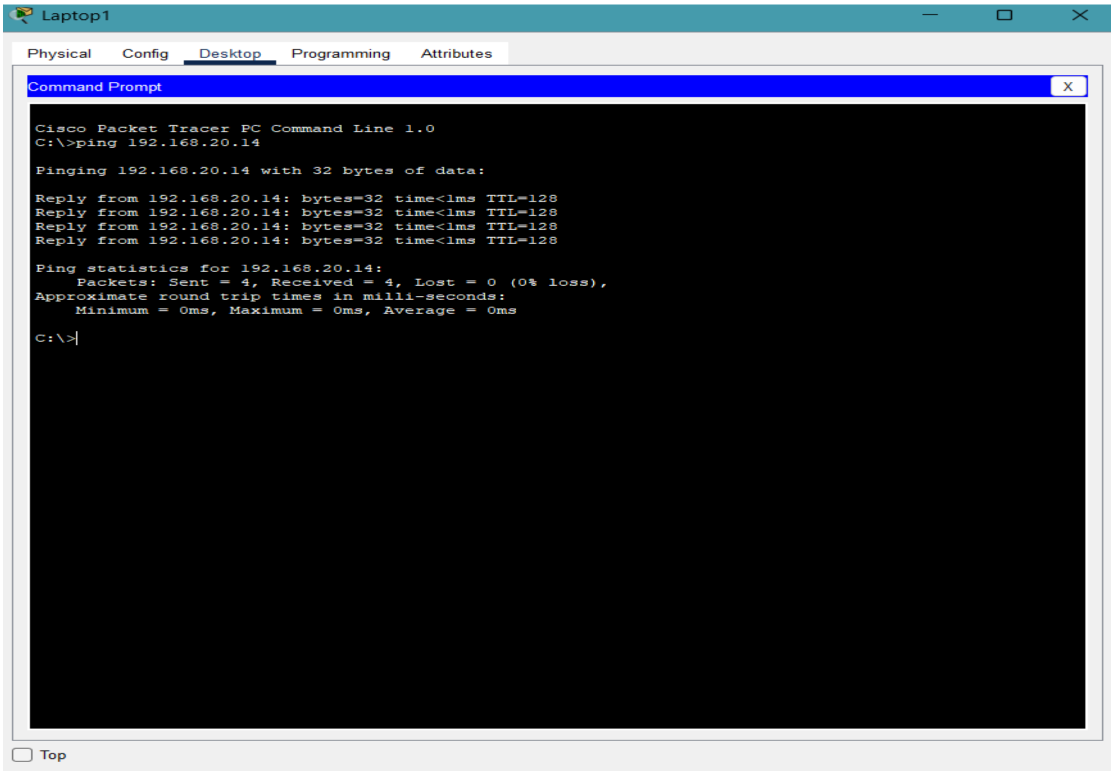
Top



Finally, I verified the configuration using 'show' commands and saved the running configuration to the startup-config to ensure all changes persist after a device reboot.



ping from Labtop1 to Pc1:



ping from Pc3 to Pc1: